BENNETT UNIVERSITY
TIMES OF INDIA GROUP

LearnExa - AI Study Companion

Presented By:
Aditya Karanwal : E23CSEU2290

Submitted To: Prashant Kumar
Professor
SCSET, Bennett University

Problem Statement

Despite the rapid advancement of Generative AI in education, there is still no unified, multimodal, voice-enabled, and document-grounded AI learning ecosystem that adapts dynamically to individual student needs.

- Most AI tools are text-based chatbots.
- Limited integration of:
    - Multimodal learning (text + image + audio)
    - Voice tutoring systems
    - Progress tracking dashboards
    - Document-grounded responses (RAG)
- Students still manually:
    - Convert PDFs to notes
    - Create quizzes
    - Track learning progress

Therefore, there exists a gap between theoretical AI capabilities and practical personalized learning systems.

Introduction

Our Purpose is to build :

An AI-powered Study Companion that transforms raw learning material into an adaptive, multimodal and personalized which make betters the users learning experience.

What We Plan to Build:

A full-stack intelligent learning platform that:

1. Converts PDFs, text, URLs into:
   - Structured summaries
   - Flashcards
   - Interactive quizzes
2. Supports:
   - Visual learning (image generation)
   - Auditory learning (Text-to-Speech)
3. Tracks:
   - Learning progress
   - Weak topics
   - Study patterns
4. Uses:
   - RAG (Retrieval-Augmented Generation)
   - Prompt-engineered role-based AI
5. Provides:
   - Grounded, personalized, adaptive tutoring

Literature Review

Paper	Focus Area	Methadology	Key Findings	Limitation
ChatISA	In-house AI chatbot for IS education	Design Science Research (DSRM), multi-model integration, Retrieval-Augmented Generation (RAG), open-source deployment	Domain-specific chatbot improves student interaction	Limited generalization, domain restriction
EDUBOT	AI-powered student assistant chatbot	GPT-4 integration, Google Custom Search API, performance visualization (Matplotlib), supervised evaluation	Improved engagement & academic performance	API dependency, limited multimodal depth
Generative AI in ITS	AI-enhanced Intelligent Tutoring Systems	Large Language Models (LLMs), automated question generation, adaptive feedback mechanisms	Dynamic question creation & personalized feedback	Pedagogical accuracy concerns, text-dominant interaction
LMFMs in Education	Large Multimodal Foundation Models	Multimodal transformer models, API integration, middleware architecture connecting AI systems	Rich multimodal interaction, enhanced engagement	Digital divide, hallucination risk, need for human oversight
Guettala et al., 2024	Generative AI in adaptive & personalized learning	Systematic literature review, case study analysis, framework-based evaluation of AI educational systems	Improved engagement, adaptive content, better learning outcomes	Ethical concerns, hallucination, lack of real-world deployment

Base Paper Overview -

Title: Generative Artificial Intelligence in Education: Advancing Adaptive and Personalized Learning

The paper explores:
- Evolution from AI → ML → DL → Generative AI
- Integration of Generative AI in adaptive learning environments
- Case studies of AI-driven educational systems
- Transition from Education 4.0 to Education 5.0

It focuses on:
- Personalized learning pathways
- Dynamic content generation
- Engagement improvement

Key Findings:
- Increased student engagement
- Improved test performance
- Accelerated skill development
- Personalized adaptive content improves retention

Limitations:
- Ethical & bias concerns
- Hallucination risks
- Lack of grounding mechanisms
- Limited multimodal real deployment
- Need for collaborative human-AI integration
- Mostly conceptual and analytical — not full implementation

Our work and Implementation

BENNETT UNIVERSITY
TIMES OF INDIA GROUP

After recognizing the limitations of previous studies, we propose a novel solution to overcome these challenges :

A Multimodal, Voice-Enabled, RAG-Grounded AI Study Companion with Full Progress Analytics

Key Modules:

1. Data Ingestion
2. RAG Retrieval System
3. Prompt Engineering Layer
4. Multimodal Output Layer
5. Progress Tracking Engine
6. Backend-Frontend Integration

Research Gap We Address:

Identified Limitation	Our Solution
Text-only systems	Multimodal learning (Text + Image + Voice)
Hallucination	RAG-based document grounding
No voice tutoring	Integrated Text-to-Speech
No personalization tracking	Database-based analytics

Workflow

Step 1: User Input
- Upload PDF / Enter URL / Enter Topic

Step 2: Data Processing
- Text extraction
- Chunking
- Embedding generation
- Store in vector database

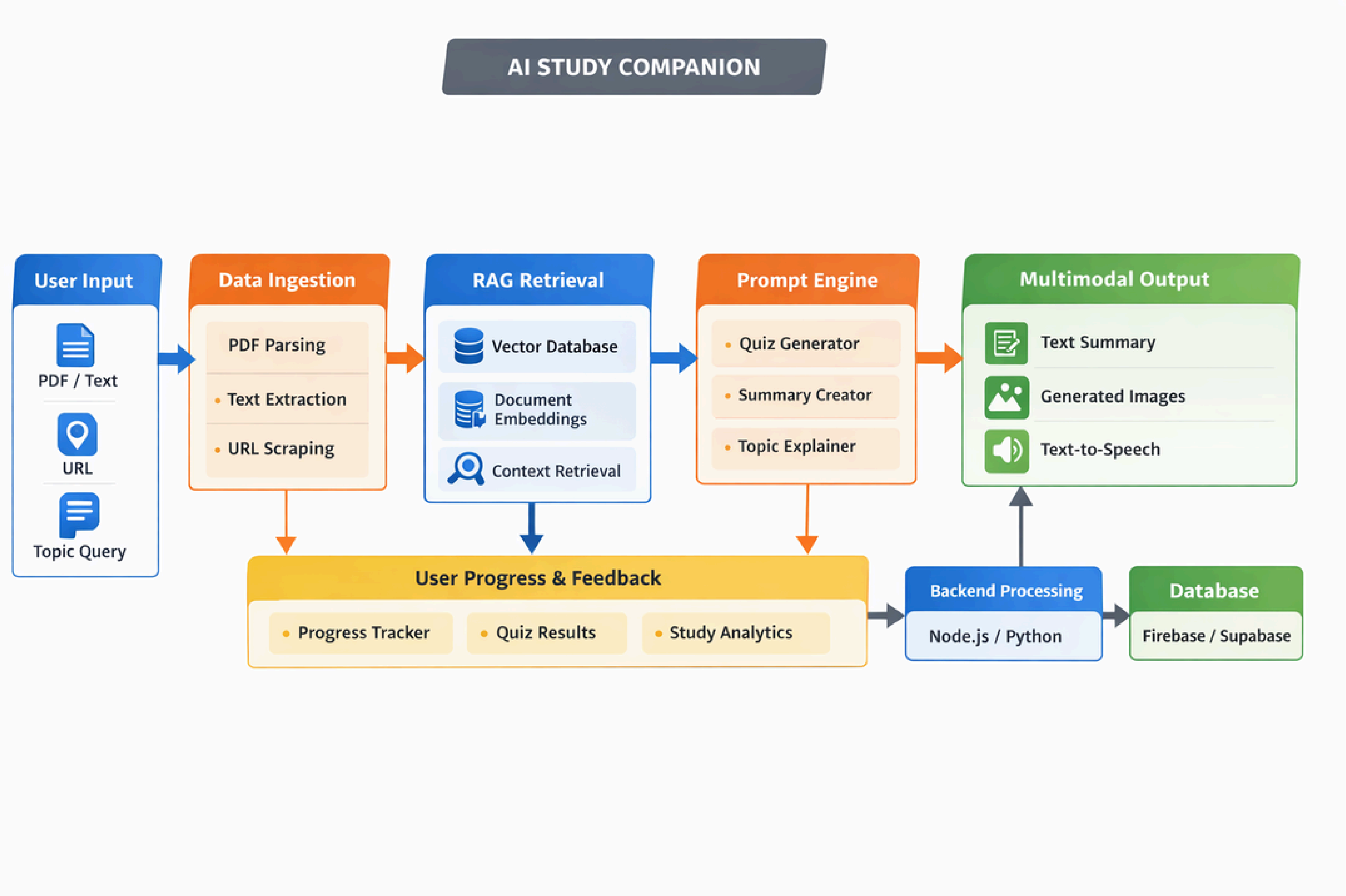
Step 3: RAG Retrieval
- Retrieve relevant document chunks
- Inject into prompt context

Step 4: Prompt Engine
- System role selection
- Generate:
  - Summary
  - Quiz
  - Explanation
  - Interview questions

Step 5: Multimodal Layer
- Generate diagram/image
- Convert explanation to voice
- Provide downloadable notes

Step 6: Progress Tracking
- Store:
  - Quiz scores
  - Study time
  - Weak topics
- Visual dashboard analytics

Step 7: Feedback Loop
- Use performance data
- Adjust difficulty
- Recommend revision topics



Methodology

1. . Model Selection
2. Uses a pretrained latent diffusion model (runwayml/stable-diffusion-v1-5 or similar).
3. Device Selection
4. Automatically chooses cuda or cpu.
5. Prompt Conditioning
6. The text prompt is encoded using the text encoder.
7. Latent Diffusion Sampling
8. The model denoises a random latent over multiple inference steps.
9. Classifier-Free Guidance
10. guidance_scale controls how strongly the image follows the prompt.
11. Decoding & Saving
12. The final latent is decoded into an image and saved.

Conclusion

This project bridges the gap between theoretical generative AI research and practical, multimodal, voice-enabled personalized learning systems.

We contribute:

- True multimodal integration
- Voice-based adaptive tutoring
- Data-driven progress tracking
- Full-stack educational deployment